

Guide: certified request cutoffs

Collatz Lean research project

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What is now checked

The earlier optimizer proposed a seed cutoff from the transfer parameters available in an induction. The new Lean checker verifies that no odd-run request below that cutoff needs an unavailable parameter.

For transfer parameters below 61, the certified seed cutoff is 415, improving the sufficient dyadic cutoff 128. For parameters below $2^{64} - 1$, the certified cutoff is 30786325577727. These are conditional convergence thresholds: all smaller transfer instances must still be available.

Using the certificate

Choose U , a proposed seed bound N , and L such that $N < 2^L$. Prove `RequestCutoff.Check U N L`. The checker considers each run length below L and the two odd residue classes modulo four. It checks only their largest eligible odd parts.

Use `RequestCutoff.reachesOne` to obtain convergence below N from all transfer instances below U . To close a transfer target that descends into this interval, use `AffineTransfer.of_request_cutoff_descent`. The source convergence argument alone does not establish the cutoff's smaller-transfer premises.

Source and verification

The formal source is `lean/Collatz/RequestCutoff.lean`. The declarations `check_61` and `check_mersenne64` are evaluated by ordinary `decide`, with no extra computational axiom. The declarations `maximal_61` and `maximal_mersenne64` exhibit the first unavailable requests and exclude every larger cutoff in the same request system.

From `lean/`, run:

```
lake build Collatz.RequestCutoff
```

From the repository root, with the pinned toolchain on `PATH`:

```
python3 analysis/audit_theorems.py --build
python3 analysis/transfer_requests.py
python3 -m unittest discover -s analysis \
  -p 'test_transfer_requests.py'
```

The Python program proposes thresholds. Only the two selected outputs described here are certified in this module; its full enumeration and optimizer are not formalized as Lean algorithms.

The remaining mathematical task

A cutoff certificate supplies an induction premise, not a descent witness. We still need a general argument that targets descend into their certified intervals, or a covering combination of this rule and other valid transfer rules. Maximality of these cutoffs does not establish nonconvergence above them. The Collatz conjecture remains unproved.